

INF142 - Computer Networks

Mandatory Assignment 2: max 18 pts

Submission Deadline: 23:59, Friday, April 17, 2026

Submission of your assignment

Instructions for your submission:

- Naming: your submission files should be named `<id>.*` where `<id>` is your StudentID. For each answer, specify which question you answer to.
- Each question counts one point. Minimum score to get the assignment approved: 12 points.
- To enhance your learning experience, please do this exercise without relying on AI tools. The exercise group sessions in weeks 15 and 16 (after Easter) may provide hints for this mandatory assignment.
- Ja, du kan svare på norsk.

1 TCP handshake & HTTP

[4 pts]

Consider the packet capture file q1.pcap.

1. Which packets (packet numbers) make up the TCP handshake with the website `https://www.skyss.no`?
2. Which packets (packet numbers) make up the TCP handshake with the website `http://www.iloveramenoodles.com`?

Focus on the HTTP protocol and answer the following questions:

3. Another website is accessed as a result of the HTTP connection to `http://www.iloveramenoodles.com`. Which one?
4. The packet trace contains less information about the connection to `https://www.skyss.no`, compared with the one to `http://www.iloveramenoodles.com`. What is the reason for this?

Explain your reasoning for each answer.

2 Wireless connection — DHCP & ARP

[6 pts]

We captured some Internet traffic into q2.pcap.

Describe in details the main steps of the process which the laptop connecting to the wi-fi hotspot went through.

Identify the following:

1. the initial broadcast message packet when the device connected.
2. the IP address that was assigned to the device.
3. the DHCP server's IP address.
4. the MAC address of the device.
5. two packets during the ARP flow for
 - IP request
 - IP announcement to the network
6. Explain the role of the DHCP server and the ARP table when a device connects to a wi-fi hotspot.

3 File downloads and packet loss

[max 8 pts]

Consider a scenario where a client connects to a network storage server to download a 25MB file. The server sent the same file twice but using different protocols. In both cases, we captured egress (sent from the server) and ingress (received by the client) traffic.

TCP

- egress: q3_TCPserv.pcap
- ingress: q3_TCPclient.pcap

UDP

- egress: q3_UDPserv.pcap
- ingress: q3_UDPclient.pcap

Answer the following questions. Please support your answers with data from the capture files.

MAC addresses

1. Consider packet number 1 in q3_TCPserv.pcap and packet number 1 in q3_TCPclient.pcap. These two frames correspond to the same transmitted datagram, so they share the same source IP address. Now compare the MAC addresses of the two frames: Are they identical? If yes, explain why, and if not, explain why not.

TCP

2. Identify one packet the client didn't receive correctly on the first attempt of transmission. How did you identify it?
3. (*Challenge: this requires some thought*) Can you think of an explanation for what has caused the packet loss event in the previous question?¹
4. How did the server respond to this packet loss event?
5. Was the file eventually correctly received? Explain your answer.

UDP

6. From the two captures, discuss the integrity of the downloaded file. Estimate the number of packets never received by the client.
7. Knowing the original file size, estimate the size of the file the client received.
8. If packets were lost, can the UDP checksum help us to reconstruct the file?

¹The answer “no”, even if it is semantically correct, will not give a point.